

Model drift = model seem good at first glance, then decrease rapidly.

Note for Lect 1 Task 2

- Pin with hashes : too slow
- Pin the base image by digest : OKay

```
# via aiohttp
zipp==4.1.0 \
--hash=sha256:25ad4e16390cd314347dd8f1de67a2ac538ae658ed4ab9db16029c07c188e97f \
--hash=sha256:4cb57381f544315db7688e976e922a2b18cdb513d21cc194eb42232ba2a3e602
# via importlib-metadata
pip-compile --generate-hashes -v 182.30s user 109.49s system 5% cpu 1:29:23.55 total
```

```
# via aiohttp
zipp==4.1.0 \
--hash=sha256:25ad4e16390cd314347dd8f1de67a2ac538ae658ed4ab9db16029c07c188e97f \
--hash=sha256:4cb57381f544315db7688e976e922a2b18cdb513d21cc194eb42232ba2a3e602
# via importlib-metadata
pip-compile --generate-hashes -v 182.30s user 109.49s system 5% cpu 1:29:23.55 total
```

```
mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaops on mainxxx
± time docker pull python:3.11-slim && docker inspect --format='{{index .RepoDigests 0}}' python:3.11-slim
3.11-slim: Pulling from library/python
Digest: sha256:9534e5a8e315485d4061ed659af0fd78a284c015f9b73661b41d6bab25604534
Status: Image is up to date for python:3.11-slim
docker.io/library/python:3.11-slim
docker pull python:3.11-slim 0.01s user 0.01s system 0% cpu 11.306 total
python@sha256:9534e5a8e315485d4061ed659af0fd78a284c015f9b73661b41d6bab25604534
```

- set seed

local → python -m src.train --n-estimators 200 --max-depth 8 --seed 20260101

Docker → config the Makefile 'SEED := 20260907'

Task 3: edit the cloudlayer/gcp.py

```
python -c "from src import config; from cloudlayer.factory import get_adapter; \
print(get_adapter(config.load()).push_image('itcs355-lab1:82df93d'))"
WARNING: Your config file at [/Users/mr.sup/.docker/config.json] contains these credential helper entries:
{
  "credHelpers": {
    "asia-southeast1-docker.pkg.dev": "gcloud",
    "asia-south1-docker.pkg.dev": "gcloud"
  }
}
Adding credentials for: asia-southeast1-docker.pkg.dev
gcloud credential helpers already registered correctly.
The push refers to repository [asia-southeast1-docker.pkg.dev/itcs355-6688132/itcs355/itcs355-lab1]
656f8225c277: Pushed
4425247be8c2: Pushed
3678bb828654: Pushed
c18af65b8842: Pushed
0531e0b9a580: Pushed
6310eb16bf42: Pushed
db840d886b65: Pushed
f9ef1b3b30d6: Pushed
c0b055422157: Pushed
7a4c3c6a84d: Pushed
b46d61a3db09: Pushed
82df93d: digest: sha256:14210a0e8fcd74ea685f90cd6b03fb9031a4b346d6830e101191a9f452f1be size: 856
asia-southeast1-docker.pkg.dev/itcs355-6688132/itcs355-lab1@sha256:14210a0e8fcd74ea685f90cd6b03fb9031a4b346d6830e101191a9f452f1be
[+] mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaops on mainxxx
```

Task 4

```
data project itcs355-6688132 was added to ABC which can be used by Google client libraries for billing and quota. Note:
[ml_ops] mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaops on mainxxx
± dvc push
Collecting
Pushing
2 files pushed
I0907 11:49:05.427763 345428 fork_posix.cc:71] Other threads are currently calling into gRPC, skipping fork() handlers
I0907 11:49:05.430227 345659 ev_poll_posix.cc:593] FD from fork parent still in poll list: fd(21, generation: 1)
I0907 11:49:05.431914 345428 fork_posix.cc:71] Other threads are currently calling into gRPC, skipping fork() handlers
[ml_ops] mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaops on mainxxx
```

It seem that source already got this, so just run

```
[ml_ops] mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaops on mainxxx
± pytest tests/test_data.py -v
platform darwin -- Python 3.11.9, pytest-9.1.1, pluggy-1.6.0 -- /Users/mr.sup/.conda/envs/ml_ops/bin/python3.11
cachedir: .pytest_cache
rootdir: /Users/mr.sup/ict1_2026/ml_ops/public_teaching_mlaops
plugins: anyio-4.15.1
collected 10 items

tests/test_data.py::test_schema_columns_present_and_typed PASSED
tests/test_data.py::test_no_null_in_required_columns PASSED
tests/test_data.py::test_features_within_plausible_ranges PASSED
tests/test_data.py::test_target_is_binary_and_not_degenerate PASSED
tests/test_data.py::test_identifier_is_unique PASSED
tests/test_data.py::test_no_machine_leaks_across_splits PASSED
tests/test_data.py::test_split_is_deterministic_given_seed PASSED
tests/test_data.py::test_split_changes_with_seed PASSED
tests/test_data.py::test_every_row_lands_in_exactly_one_split PASSED
tests/test_data.py::test_data_fingerprint_is_stable PASSED

===== 10 passed in 0.11s =====
```

```
(ml_ops) ~mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaiops on main***  
└─ make verify  
python scripts/verify_metric.py  
claimed 0.8480 ± 0.0100  
actual 0.8482  
delta 0.0002
```

```
└─ make test  
pytest -q tests/  
.....  
===== warnings summary =====  
../../../../.conda/envs/ml_ops/lib/python3.11/site-packages/starlette/testclient.py:53  
/Users/mr.sup/.conda/envs/ml_ops/lib/python3.11/site-packages/starlette/testclient.py:53: DeprecationWarning: The anyio.abc.BlockingPortal alias is deprecated, use anyio.from_thread.BlockingPortal instead.  
_PortalFactoryType = Callable[[], AbstractContextManager[anyio.abc.BlockingPortal]]  
  
-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html  
33 passed, 1 warning in 1.82s
```

Sep 07 2569 Mon
17:37
Screenshot 2569-07...3.22.44 2parrots.jpeg [100%]
Screenshot 2569-07...6.58.29 scl_figure_confer nce.png

```
(ml_ops) ~mr.sup at Suppanats-MacBook-Pro in ~/ict1_2026/ml_ops/public_teaching_mlaiops on main***  
└─ make portability-audit  
python scripts/portability_audit.py  
PORTABILITY AUDIT PASSED - src, service, monitoring, tests contain no provider-specific strings
```